

Mathematics II

CSIT — 2nd Semester — 2080 — Answer Sheet
Subject Code: MTH168

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What is a system of linear equations? When the system is consistent? Find a condition on (g) , (h) , (k) that makes the system consistent $[x_1 - 4x_2 + 7x_3 = g]$ $[3x_2 - 5x_3 = h]$ $[-2x_1 + 5x_2 - 9x_3 = k]$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Let $A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}$, $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$ (a). Compute $T(u)$ (b). Find a vector $x \in \mathbb{R}^3$ whose image under T is b (c) Is x unique?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Find a least square solution of $Ax = b$ where $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ -3 \\ -5 \end{bmatrix}$. Also, compute the associated least square error.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Are the vectors $v_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} -5 \\ 0 \\ 6 \end{bmatrix}$ linearly independent? Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find LU factorization of $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 5 & 10 \\ 4 & 8 & 2 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Compute $\det(A)$ where $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ -4 & 0 & 6 & 6 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Show that $H = \{(a - 3b, b - a, a, b) : a, b \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^4$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Is $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 5 & -3 \\ -4 & 9 \end{bmatrix}$? If so, find the eigenvalue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Let $u = (1, -2, 2, 0)$. Find a unit vector v in the same direction as u .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Find the basis and dimension of $\text{Nul } A$ where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define group. Show that $(\mathbb{Z}, +)$ doesn't form a group.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Show that every field is an integral domain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.